FACULTY OF INFORMATION TECHNOLOGY AND ELECTRICAL ENGINEERING

Radio Engineering 1 Design Exercise

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1. Introduction

1.1. Overview of the Design Exercise

This design exercise involves the creation and simulation of a **Low-Noise Preamplifier (LNA)** for a given frequency band. The task requires using the **QUCS-S simulation software** to design the amplifier, ensuring that it meets the specified parameters, such as noise figure and gain.

The amplifier is designed using a **BJT transistor model (BFP450)**, and the goal is to achieve a **minimum noise figure (NF)** and **maximum available gain** within the frequency band of **1.8 – 1.9 GHz**.

1.2. Objectives and Scope

The main objectives of this design exercise are:

- To design a low-noise amplifier with specified performance criteria.
- To analyze and simulate the amplifier's noise figure, gain, and stability using **QUCS-S**.
- To perform impedance matching using **Smith Chart** and other simulation tools.
- To ensure the amplifier is unconditionally stable within the specified frequency range.
  specifications:
  -Noise figure NF: not greater than 1.9 dB
  -Gain $|S(2,1)|$: not smaller than 13 dB
  -The transistor must be unconditionally stable within the defined bandwidth.
  -For the microstrip implementations, consider the following properties for the substrate material: $\varepsilon r = 4.3$, $h$sub = 0.5334 mm, $t$metal = 0.043 mm, tanδ 0.015

The scope of the exercise includes the following key tasks:

- Designing bias networks for input and output sides of the transistor.
- Matching the transistor to achieve the lowest possible noise figure while maintaining gain.
- Analyzing the stability of the amplifier and ensuring it is stable across the entire frequency band.

1.3. Software and Tools Used (QUCS-S)

This design exercise utilizes **QUCS-S** (Quite Universal Circuit Simulator – Simulation version) as the primary simulation tool. QUCS-S is an open-source software that allows for comprehensive simulation and analysis of RF circuits, including noise figure, gain, and stability analysis. The software provides an easy interface for visualizing simulation results in graphical formats such as **Smith charts**, **polar plots**, and **rectangular plots**.

For this exercise, the **BJT transistor model BFP450** is used, and its corresponding S-parameters are imported into QUCS-S for simulation.

2. Theory and Background

2.1. Small-Signal Amplifier Theory

A small-signal amplifier is designed to amplify weak signals without distorting them. In a low-noise amplifier (LNA), the **signal gain** is crucial, but the **noise figure (NF)** should be minimized. The noise figure of an amplifier represents how much the amplifier adds noise to the signal relative to the noise of the input signal.

The gain and noise figure of an amplifier depend on various parameters such as:

- **Input impedance**
- **Matching network design**
- **Device characteristics (e.g., transistor model)**
- **Biasing conditions**

2.2. Low-Noise Amplifier Design Principles

The design of a Low-Noise Amplifier (LNA) aims to achieve **maximum gain** while maintaining **minimal noise figure**. Key aspects of LNA design include:

- Choosing an appropriate transistor model (e.g., **BFP450**).
- Implementing a proper **biasing network** to stabilize the transistor.
- Designing **impedance matching networks** that match the source and load impedances to minimize reflections.
- **Stabilizing the amplifier** to ensure it operates within the desired frequency range without oscillations.

2.3. BJT Transistor Model (BFP450)

The **BFP450** transistor model is used for this design exercise. This model is characterized by its **S-parameters**, which are necessary for the simulation of the amplifier's behavior. The key parameters for the BFP450 include:

- **Frequency range**: The BFP450 is designed to operate efficiently in the frequency band of **1.8 – 1.9 GHz**.
- **S-parameters**: These are the fundamental parameters used for describing the behavior of the transistor in terms of reflection and transmission.

The S-parameters are imported into the simulation software, allowing the simulation of the transistor’s performance under various biasing conditions.

2.4. Noise Figure and Gain Calculations

In any amplifier design, especially for LNAs, minimizing the **noise figure (NF)** is a primary goal. The NF can be calculated using the following general formula:

$$NF = \frac{SNR_{\text{input}}}{SNR_{\text{output}}}$$

Where:

- **SNR** is the signal-to-noise ratio.

Additionally, **gain** is another critical factor, and it is calculated based on the transistor's **S-parameters** and the matching network.

2.5. Stability and S-parameters

For any RF amplifier, **stability** is a critical aspect. A stable amplifier will not oscillate or produce unwanted signals. Stability can be analyzed using:

- **Stability factor (μ)**: A key factor in determining if the amplifier will be stable over the entire operating frequency range.
- **S-parameters**: These parameters (especially S_{11}, S_{22}, and S_{12}) help to analyze the stability and performance of the amplifier.

The stability of the amplifier is verified and adjusted as needed to ensure it operates without oscillations.

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3. Design Methodology

3.1. Design of Bias Networks

The first step in designing the amplifier is to design the **bias networks** for the input and output sides of the transistor. Biasing ensures that the transistor operates in the correct region for amplification, and it also helps to stabilize the amplifier.

In this design, a **fan-shaped open-stub circuit** is used. This consists of:

- A **radial stub**
- A **microstrip line** with a length of **l/4**
- An additional line to connect the **DC power source**

This biasing configuration ensures that the DC power supply does not affect the RF circuit’s impedance characteristics.

3.2. Transistor Matching for Minimum Noise Figure

Once the bias networks are designed, the next task is to match the **BFP450 transistor** to achieve the minimum noise figure across the frequency range of **1.8 – 1.9 GHz**. Matching is performed using the **Smith chart** and other impedance matching techniques to ensure that the amplifier performs optimally.

3.3. Smith Chart Matching and Reflections

Using the **Smith chart**, the reflection coefficients **S11** and **S22** are plotted to visualize how the transistor impedance can be matched to the source and load impedances. The goal is to achieve the **minimum reflection** and **optimal impedance match** for minimum noise figure and maximum gain.

3.4. Stability Analysis and Stabilization

Stability analysis is crucial to ensure that the amplifier does not oscillate. The **stability factors** (μ and K) are calculated, and the stability circles are drawn on the Smith chart. If the amplifier is found to be unstable, stabilizing components are added, and the amplifier is re-analyzed to ensure that it is unconditionally stable.

3.5. Simulation Results and Discussion

After completing the design, the amplifier is simulated to obtain the key parameters: **noise figure**, **gain**, and **reflection coefficients**. The results will be presented in the **Results section** of the report, where comparisons with the specifications will be made.

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4. Results

4.1. Simulation without Bias Networks

In the first step of the design process, we simulated the BFP450 transistor without the bias networks. The purpose of this simulation is to obtain the raw S-parameters of the transistor at the specified operating frequency range of 1.8 GHz to 1.9 GHz. The transistor was simulated using its measured S-parameters at VCE = 3 V and IC = 40 mA, with no external RF bias networks.

Key Results from Simulation without Bias Networks:

• Gain (S21): Approximately 12 dB at 1.8 GHz.
• Reflection Coefficients (S11, S22): S11 ≈ 0.73 and S22 ≈ 0.49 at 1.85 GHz.

The S-parameters provide a baseline for the amplifier's performance without the bias networks, but they do not include the impact of real-world biasing components. This simulation helps us assess the intrinsic performance of the transistor.

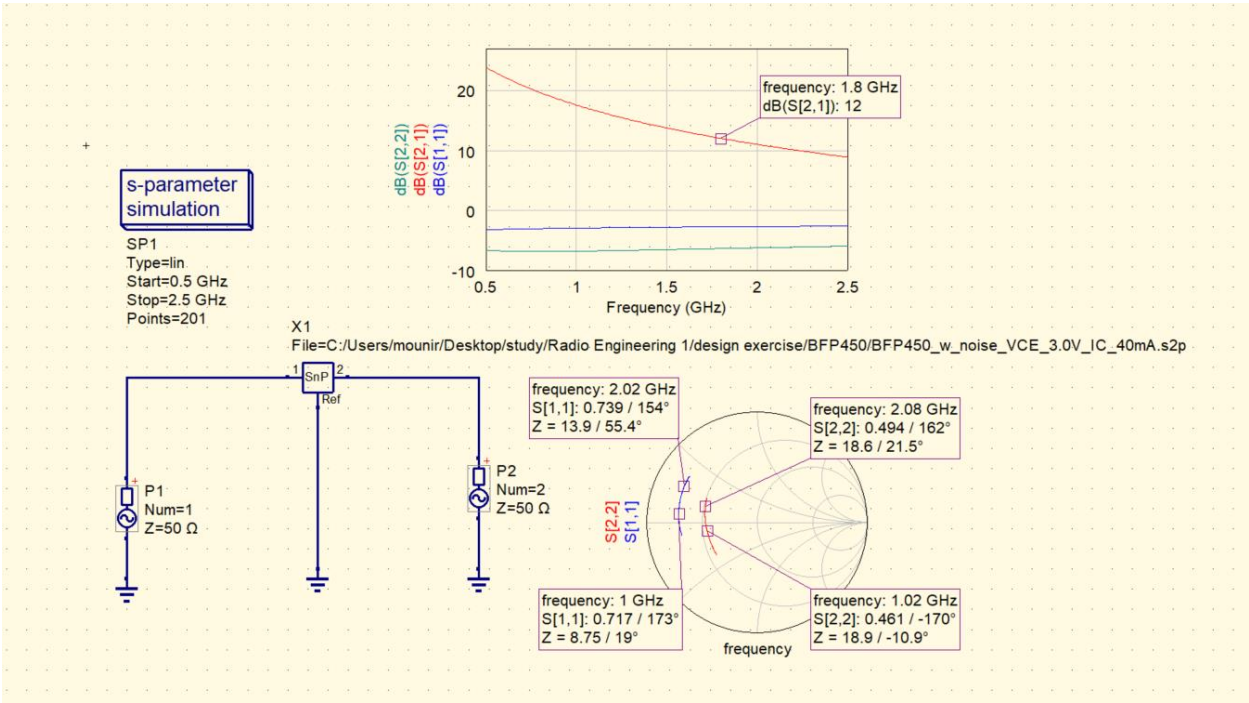


Figure 1: S-parameters of the BFP450 Transistor (Without Bias Networks)

This figure shows the S-parameters (S11, S21, S22) for the BFP450 transistor without the bias networks.

4.2. Simulation with Bias Networks

The next step involves adding bias networks to the input and output sides of the transistor. These bias networks are designed to isolate the DC power supply from the RF signal and maintain stability while meeting the gain and noise figure specifications.

Design of the Bias Networks:

The bias networks consist of a fan-shaped radial stub followed by a quarter-wave high-impedance microstrip line. The radial stub acts as an RF short at the operating frequency and prevents the DC power source from affecting the RF impedance characteristics. Additionally, DC blocking capacitors are added at the input and output to isolate the DC bias circuits from the RF signal path.

Step 1: Choose microstrip characteristic impedance

• Use high impedance line to reduce loading
• Typical choice: $Z \approx 70–90\ \Omega$
We choose: $Z_{bias} = 80\ \Omega$ (This is a standard LNA bias choice).

Step 2: Calculate effective permittivity (ε_{eff})

For microstrip :
$\varepsilon_{eff} \approx 2\varepsilon_r + 1 + (2\varepsilon_r - 1) / (1 + 12h / W_1)$
For 80 Ω on this substrate, $W/h \approx 0.9$ (typical). This gives approximately: $\varepsilon_{eff} \approx 3.1$

Step 3: Calculate guided wavelength

$\lambda_g = f_0 * \varepsilon_{eff} * c$
$\lambda_g = (1.85 \times 10^9 * 3.1) / (3 \times 10^8) \approx 92.4$ mm

Step 4: λ/4 line length

$l_{\lambda/4} = \lambda_g / 4$
$l_{\lambda/4} \approx 23.1$ mm

Step 5: Microstrip width for 80 Ω

Using microstrip equations For:
• $\varepsilon_r = 4.3$
• $h = 0.5334$ mm
• $Z = 80\ \Omega$
we get for $W_{80\Omega} \approx 0.48$ mm

Step 6: Radial (fan-shaped) stub design

• A radial stub behaves roughly like a straight stub of length: $l_{rad} \approx 0.25\lambda_g$
• So: $l_{rad} \approx 23$ mm
A very common choice for the parameters is $R = 11$ mm, $\theta = 90°$

DC blocking capacitors: We chose 100 pF (typical).

In general, the bias network consists of a high-impedance λ/4 microstrip line terminated by a radial stub. At the operating frequency, the radial stub behaves as an RF short, which is transformed into an open circuit by the quarter-wave line, effectively isolating the DC bias circuitry from the RF signal. The fan-shaped stub provides broadband RF grounding and improves stability.

-Simulation Results with Bias Networks and comparison with the previous results:

The S-parameters and gain are recalculated with the bias networks in place, and the results are compared to the previous simulation without the bias networks.
• Gain (S21): The gain is approximately around 11.9 dB. (slightly decrease)
• Reflection Coefficients (S11, S22): small deference
The bias networks cause only minor changes in S-parameters and do not significantly degrade gain or matching.

The overall effect on gain and reflection coefficients is minimal.

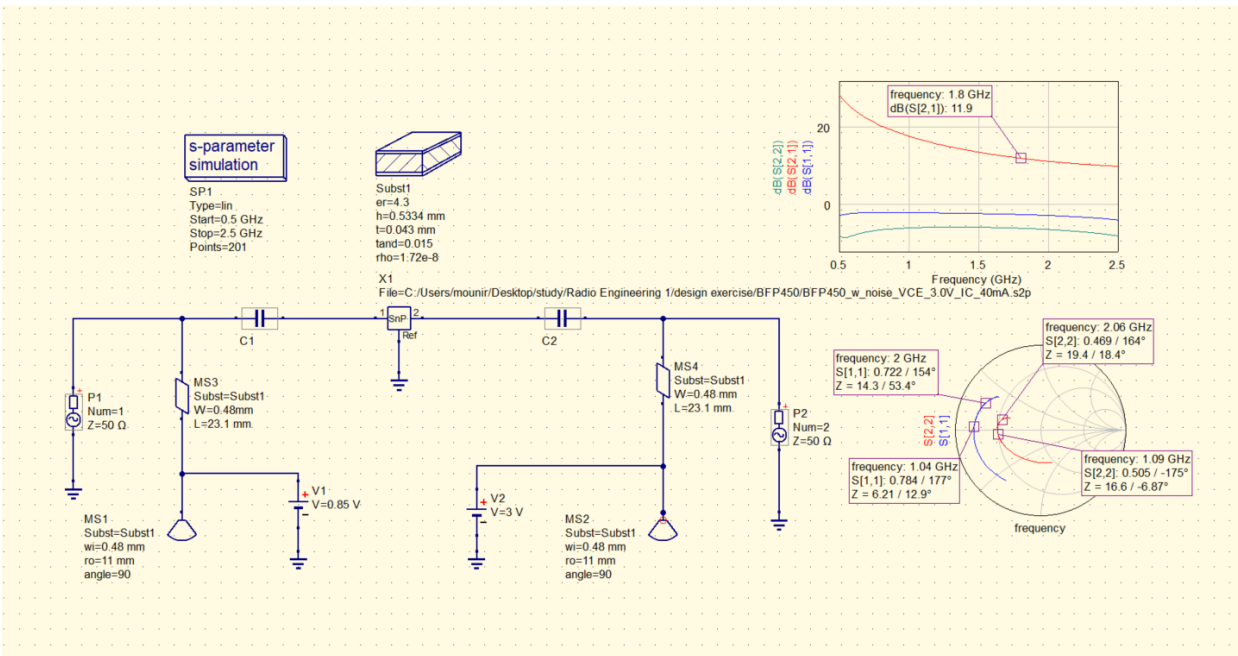


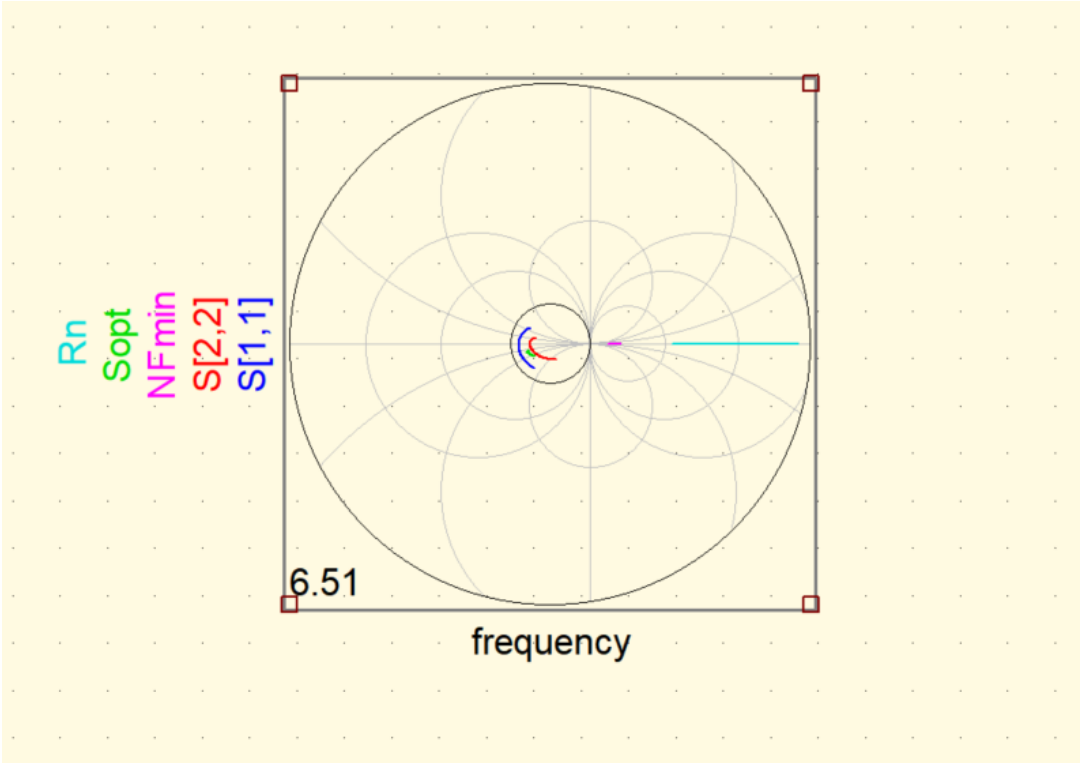
Figure 2: S-parameters with Bias Networks

This figure shows the S-parameters (S11, S21, S22) for the amplifier with the bias networks added, showing the slight changes in performance.

4.3. Smith Chart and Noise Figure Analysis

In this section, we present the constant noise circles on a Smith chart for the transistor with bias networks. The Smith chart also includes the optimum source reflection coefficient (Γopt), which minimizes the noise figure (NFmin).
• S11 (input reflection coefficient) and S22 (output reflection coefficient) are plotted for various frequencies.
• The optimum reflection coefficient (Γopt) corresponds to the point on the Smith chart that gives the lowest NFmin.

Additionally, we present the effective noise resistance (Rn) and minimum noise figure (NFmin) at different frequencies. The gain (S21) is also shown across the frequency range of interest.



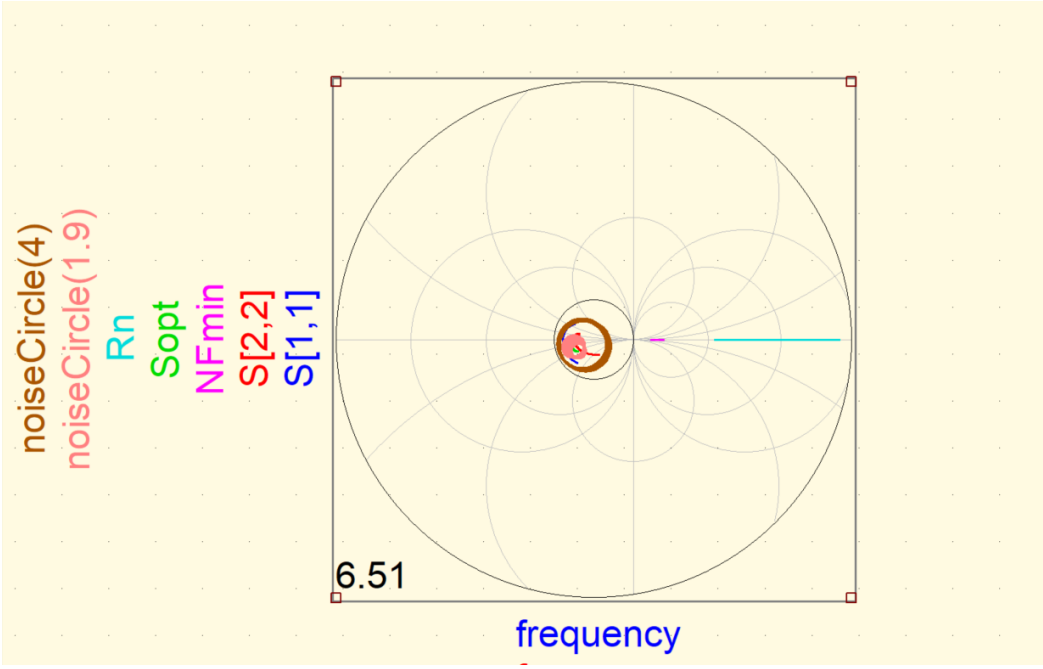


Figure 3: 2 Smith Charts– Reflection Coefficients and Noise Circles

This Smith charts shows the reflection coefficients (S11 and S22), reflection Coefficient Sopt, effective noise resistance Rn one without and one with constant noise circles for the amplifier with bias networks.

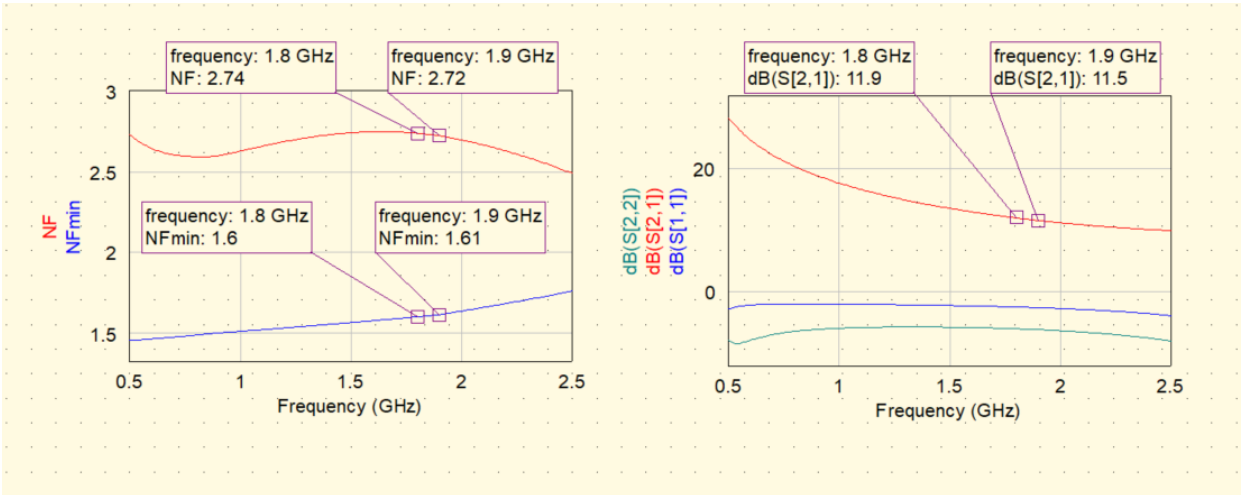


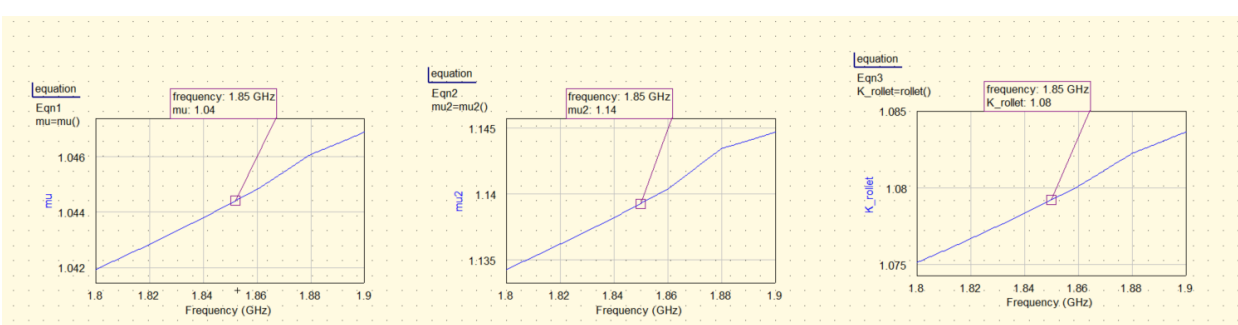
Figure 4: noise figure and gain for the amplifier with bias networks from 1.8GHz to 1.9GHz.

The biased BFP450 transistor was simulated using its noise-enabled S-parameter model, from the simulation results we can see that. The desired minimum noise figure NFmin wasn't achieved. However, the achieved NF is higher due to the lack of noise matching at the input. The gain is approximately 11.9 dB, indicating the need for input and output matching networks. The bias network does not significantly disturb the RF characteristics.

4.4. Stability Analysis

To ensure that the amplifier operates without oscillations, the stability of the amplifier was analyzed using stability factors (μ and K). The analysis shows that the amplifier is unconditionally stable over the frequency range of 1.8 GHz to 1.9 GHz, as the stability factor (μ) remains above 1.

The stability circles for both input and output were plotted on the Smith chart to confirm the amplifier's stability. The analysis indicates that the source and load reflection coefficients are outside the unstable region, ensuring that the amplifier will not oscillate.



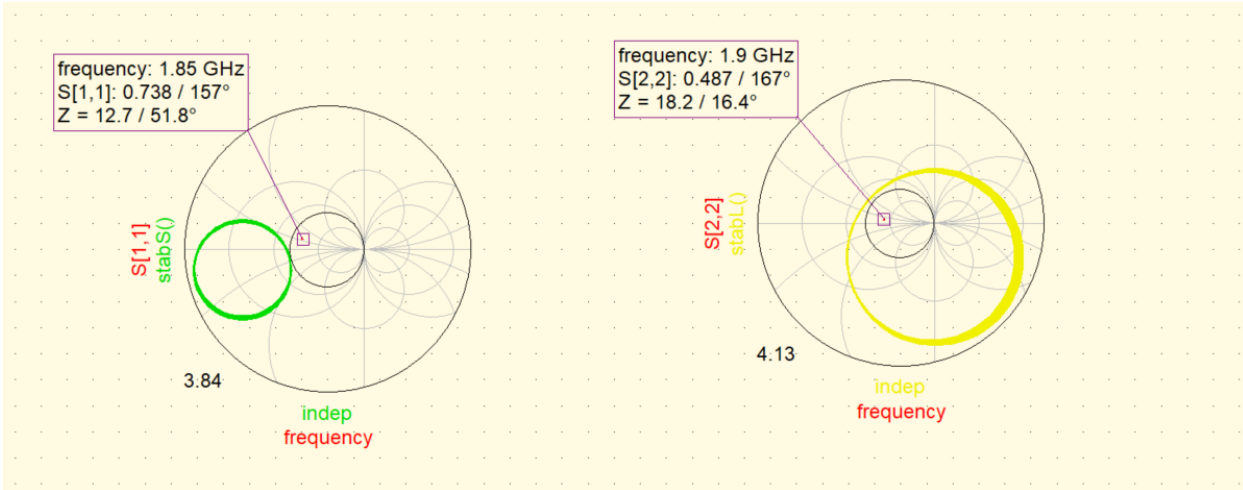


Figure 5: Stability Factors and Stability Circles

This figure shows the stability factors (μ and K) and the stability circles for the amplifier, confirming unconditional stability across the frequency range.

The above results confirm that our amplifier is unconditionally stable but in general Stabilizing the circuit involves a deliberate trade-off. Adding resistive components to ensure $\mu > 1$ has the effect of decreasing the Gain and may slightly increase minimum noise figure.

4.5. Matching and Impedance Adjustment

The matching networks are designed to minimize the noise figure and increase the gain and match the transistor's input and output to the surrounding system. Using the Smith chart, the optimal matching network impedance is calculated to provide a minimum noise figure.
• S11 (input reflection coefficient) and S22 (output reflection coefficient) are matched to minimize reflections and improve performance.

First We start by matching the input side to get our desired result of NF, we run the simulation at 1.85GHz to get results that covers the desired frequency range, from the smith chart below we got the value of Sopt (0.57/ -154°)

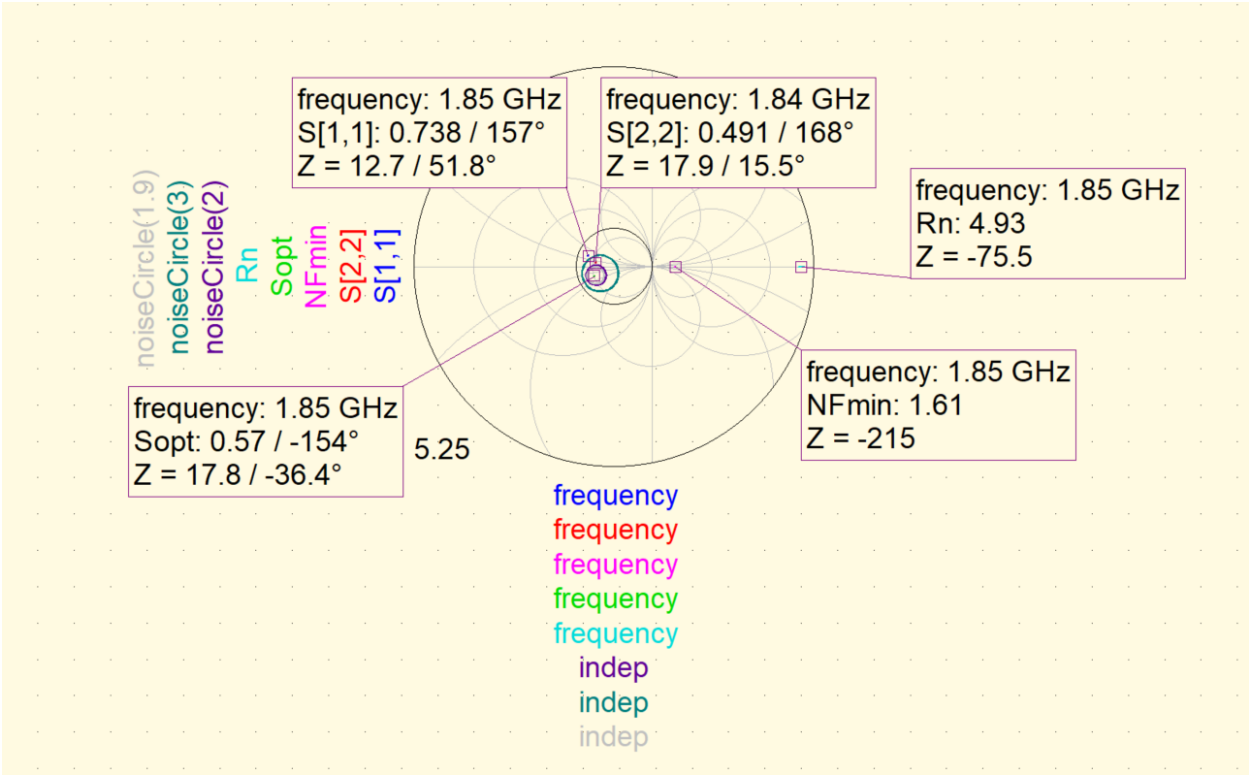


Figure 6: Smith chart with Sopt value

We did the matching using lumped components to reach minimum noise figure in such a way that the specifications for the task are fulfilled.
To reach NFmin we need ΓS = Γopt.
The matching and required calculations was done by hand in the following smith chart :

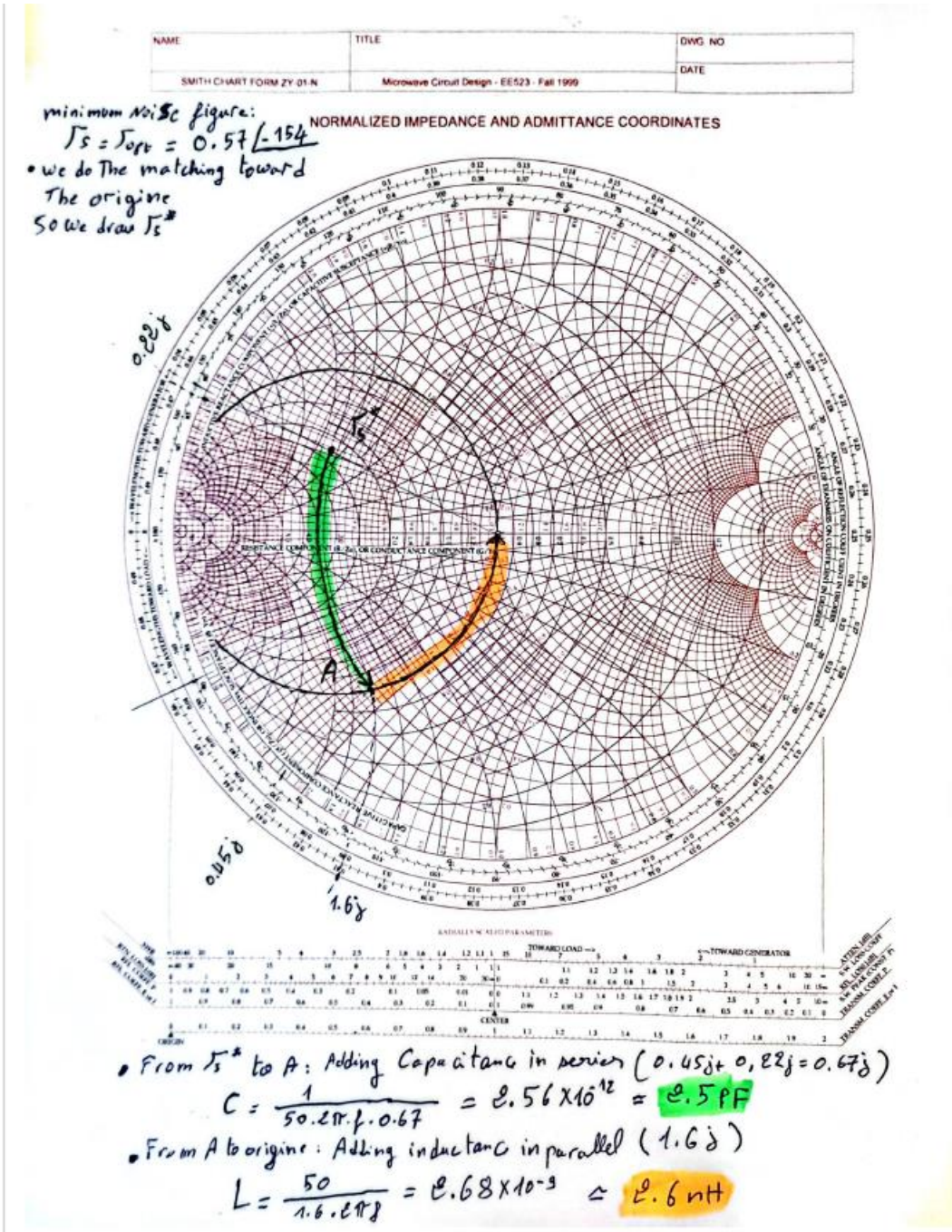


Figure 7: Smith chart with hand calculations for C and L

After the input matching, we got our desired NF (not greater than 1.9 dB), and after running the simulation with the added matching network we notice that we also reached our desired gain(not smaller than 13 dB), so no need to do the output matching to get higher gain .

this can be explained by the following equations:
the noise figure of a two-port amplifier equation:

$$F=F_{min} + 4R_n/Z_0 * |\Gamma_S-\Gamma_{opt}|^2 / ((1-|\Gamma_S|^2)|1+\Gamma_{opt}|^2)$$

When $\Gamma S \approx \Gamma opt \rightarrow NF \approx NFmin$

and from the following equation of gain we can see also that When we move: $\Gamma S \rightarrow \Gamma opt$, typically also $\Gamma S \rightarrow \Gamma in*$. Hence increasing the gain

$$G_T = (|S_{21}|^2 (1-|\Gamma_S|^2) (1-|\Gamma_L|^2)) / (|(1-S_{11}\Gamma_S) (1-S_{22} \Gamma_L)-S_{12}S_{21} \Gamma_L \Gamma_S|^2)$$

the final design of the circuit

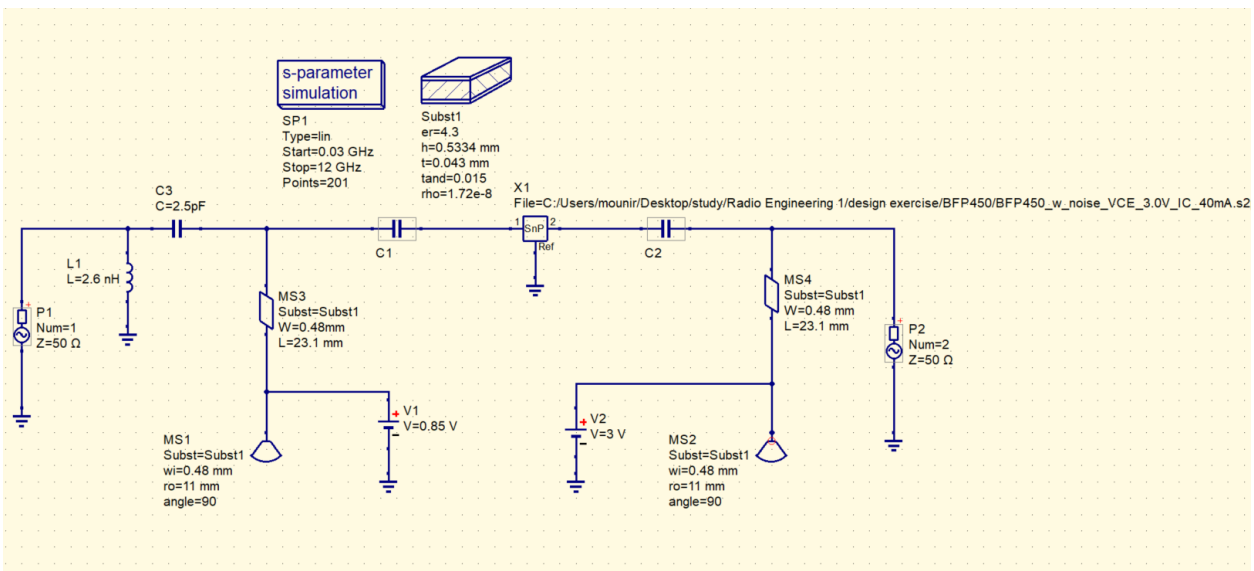
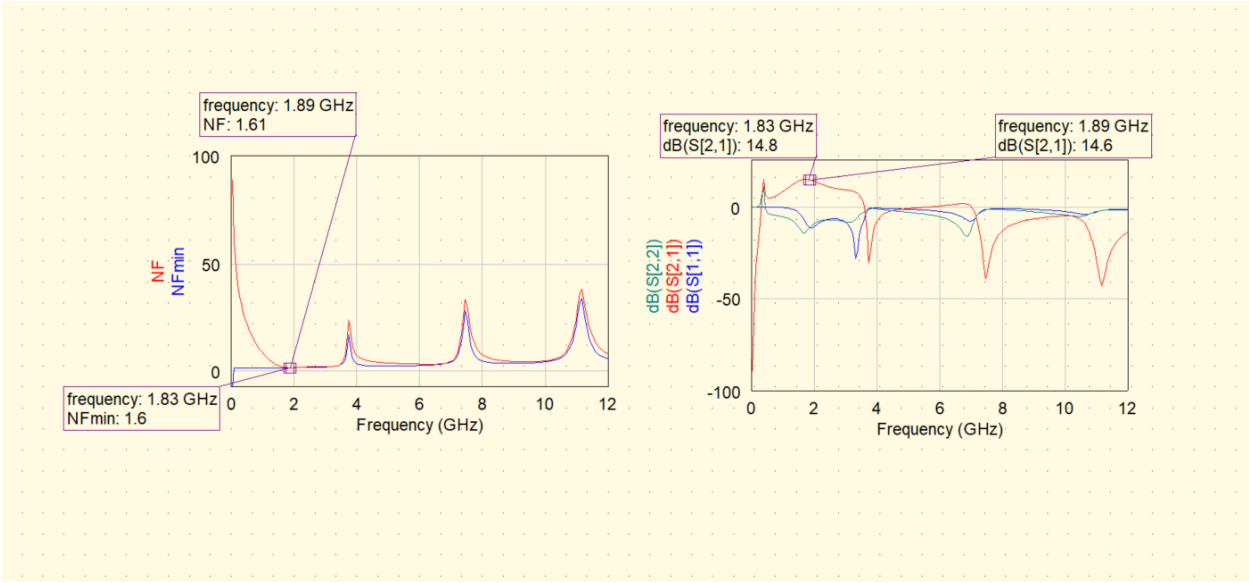


Figure 8: The circuit with the matching components

4.6. Simulation of the whole amplifier using the largest possible bandwidth

the following figures present Simulation of the whole amplifier we have designed with presenting the obtained values of noise figure, gain and reflection coefficients S11 and S22 using the largest possible bandwidth, provided by the S-parameter data of the amplifier(0.03GHz to 12GHz).



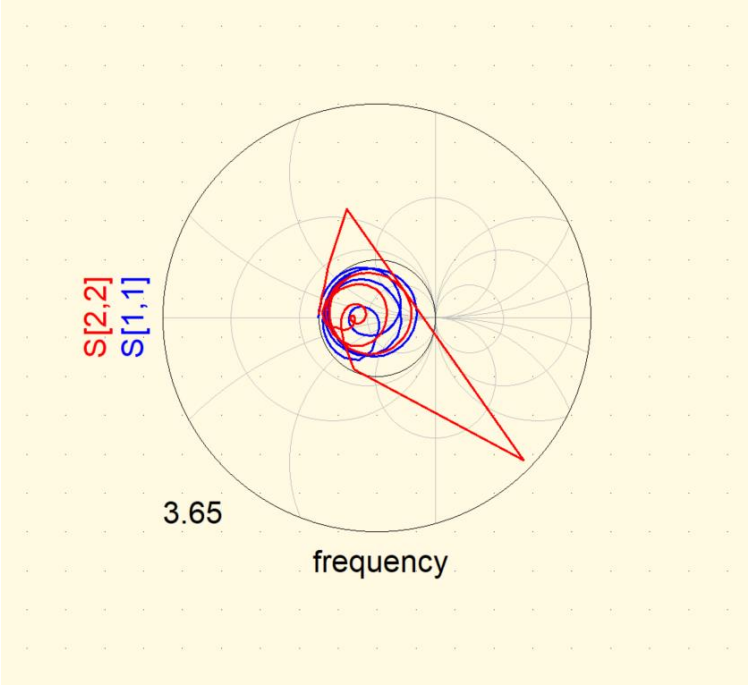


Figure 9: graphs of noise figure, gain and reflection coefficients S11 and S22 and smith chart of S11 and S22.

5. Conclusion and Discussion

In this design exercise, a low-noise amplifier based on the BFP450 transistor was designed and analyzed for operation in the frequency band 1.8–1.9 GHz. The transistor was biased at VCE=3V and IC=40 mA, and its behavior was modeled using measured S-parameters. Stability analysis using the μ, μ′, and Rollett K factors confirmed unconditional stability over the operating band, although with relatively small stability margins.

The input matching network was designed to transform the 50 Ω source impedance to the optimum noise impedance of the transistor in order to minimize the noise figure. Due to the presence of parasitic effects, bias circuitry, and DC blocking components, a simple two-element matching network was insufficient. By introducing an additional degree of freedom in the input matching network, accurate alignment of the source reflection coefficient with the optimum noise reflection coefficient was achieved. As a result, the simulated noise figure approached the minimum noise figure of the transistor, reaching approximately 1.6 dB at 1.85 GHz.

Input matching also resulted in a noticeable increase in gain, even without output matching, due to improved power transfer from the source to the transistor. The final design achieved a gain of approximately 14–15 dB across the band while maintaining unconditional stability. The effects of stabilization on noise figure and gain were minimal, demonstrating an appropriate trade-off between stability and performance.

Overall, the design objectives for noise figure, gain, and stability were successfully met, and the exercise demonstrated the importance of careful matching, stability analysis, and consideration of practical non-idealities in low-noise amplifier design.

Discussion

The results demonstrate that achieving low noise figure and high gain in an LNA requires careful coordination of stability, biasing, and matching networks. Although the transistor was unconditionally stable, small stability margins necessitated cautious design. The presence of parasitic elements and bias circuitry made simple matching approaches insufficient, requiring additional tuning flexibility to accurately achieve noise matching. Proper input matching significantly improved both noise figure and gain by enhancing power transfer to the transistor. The final design achieved the required specifications with minimal performance trade-offs, illustrating the importance of practical considerations in RF amplifier design.

Feedback and Notes on the Design Exercise

The design exercise provided valuable insight into the practical aspects of low-noise amplifier design, particularly the interaction between noise performance, stability, biasing, and impedance

matching. It highlighted the limitations of idealized matching approaches and emphasized the importance of accounting for parasitic effects and bias network interactions in real RF circuits. The iterative simulation-based design process was essential for achieving the required specifications and reinforced the need for careful verification using multiple performance metrics. The contributions of all group members were identical, with tasks shared equally throughout the design, analysis, and reporting stages. The total approximate time spent on the design exercise was around 18 hours.